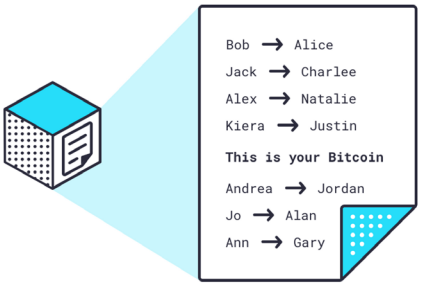
Multiple miners can be working on creating new blocks simultaneously. The first miner to discover a block hash that meets the target set by the network announces it to the network, and the other miners then **check** the transactions in that miner's candidate block to make sure they are valid. If the transactions are indeed valid, the block is added to the blockchain. The other blocks created by the other miners at the time are not added and are discarded. This process helps maintain consensus within the network and prevents double-spending.

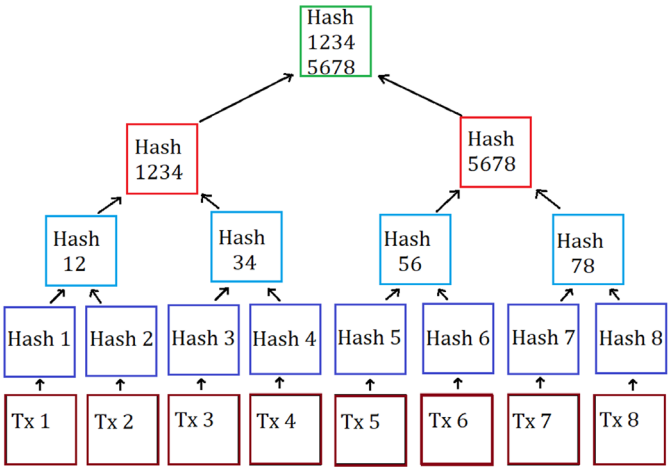
A **candidate block** is a block of transactions that is being considered for addition to the blockchain but has not yet been added.

8.3 Dissecting the Block

A *blockchain* is made up of blocks, similar to pages in a ledger, that store new **transactions**. Each block has a **header** with a summary of the data, a link to the previous block, and a unique number called a **nonce**, or a **number used once** and a few other details. Miners' task is to correctly complete the header information when creating candidate blocks.



The Organization of Transactions

Miners must arrange the transactions in their candidate blocks in a specific format, where only some of the information is included in the header.

Transactions form the backbone of the ***Bitcoin Network*** and are efficiently and securely organized through the use of **Merkle Trees**. These trees condense large amounts of data into a compact representation, improving the overall security and efficiency of the network.